Topic: Perfectly Competitive Markets, Monopoly & Monopsony

Review: Cost Concepts

Q: What are fixed costs?
A: Fixed costs ($F \geq 0$) are costs that are independent of the output quantity (q).

Q: What are variable costs?
A: Variable costs ($VC(q) \geq 0$) are costs that depend on the output quantity (q).

Q: How is total cost ($C(q)$) defined in terms of fixed and variable costs?
A: Total cost is defined as:

$$C(q) = F + VC(q)$$

Q: How is marginal cost (MC) derived from the total cost function?
A: Marginal cost is the derivative of the total cost (or variable cost) function with respect to quantity:

$$MC(q) = C'(q) = VC'(q)$$

Q: What are average cost (AC) and average variable cost (AVC)?
A: They are defined as:

$$AC(q) = \frac{C(q)}{q}, \quad AVC(q) = \frac{VC(q)}{q}$$

> **Note**
>
> The marginal cost (MC) curve intersects both the average cost (AC) and average variable cost (AVC) curves at their minimum points.

Q: How can fixed costs be decomposed?
A: Fixed costs can be decomposed into **sunk costs** (S) and **avoidable fixed costs** ($F - S$).

Q: What is the total cost function when accounting for sunk and avoidable fixed costs?
A: The cost function is piecewise:

$$C(q) = \begin{cases} S & \text{if } q = 0 \\ F + VC(q) & \text{if } q > 0 \end{cases}$$

> **Note**
>
> In this function, S represents sunk costs ($S \leq F$), and $F - S$ represents avoidable fixed costs.

Perfect Competition

Q: What are the six main assumptions of a perfectly competitive market?
A:

1. Homogeneous goods
2. Perfectly divisible output
3. Perfect information
4. No transaction costs
5. No externalities

6. Price-taking behavior

> ⚠ **Warning**
>
> Perfect competition is a theoretical benchmark and is rarely observed in reality.

Q: What is price-taking behavior?
A: It is the assumption that a firm's own output choice does not affect the prevailing market price.

Q: What is the profit function of a competitive firm?
A: The profit function is:

$$\pi(q) = R(q) - C(q) = p \cdot q - C(q)$$

Q: What are the two steps in a competitive firm's production decision process?
A:

1. Maximize profit for a positive output ($q > 0$).
2. Check if producing this optimal output is better than shutting down ($q = 0$).

Q: What is the first-order condition (FOC) for a competitive firm's profit maximization?
A: The FOC is that price equals marginal cost:

$$p = MC(q^*)$$

Q: What is the second-order condition (SOC) for a competitive firm's profit maximization?
A: The SOC requires that marginal cost be non-decreasing at the optimum:

$$MC'(q^*) \geq 0$$

> **Note**
>
> This implies marginal costs must be increasing at the profit-maximizing quantity q^*.

Q: What is the general shutdown condition for a firm?
A: A firm will produce only if the profit from producing is at least as high as the profit from shutting down:

$$\pi(q^*) \geq \pi(0) \Rightarrow p \geq AVC(q^*) + \frac{F - S}{q^*}$$

> **Note**
>
> - If all fixed costs are sunk ($S = F$), the condition simplifies to $p \geq AVC(q^*)$.
> - If no costs are sunk ($S = 0$), the condition simplifies to $p \geq AC(q^*)$.

Q: How is a firm's short-run supply function $s(p)$ defined?
A: The supply function is defined piecewise, considering the shutdown price p_s:

$$s(p) = \begin{cases} 0 & \text{if } p < p_s \\ 0 \text{ or } q_s & \text{if } p = p_s \\ q^* \text{ where } MC(q^*) = p & \text{if } p > p_s \end{cases}$$

> **Tip**
>
> The shutdown price p_s can vary between the minimum of the AVC and AC curves.

Q: Given the cost function $C(q) = \begin{cases} 12 & q = 0 \\ 16 + q^2 & q > 0 \end{cases}$, what is the firm's supply function?

A:

1. Find the optimal quantity: $MC(q) = 2q$, so $p = 2q$ implies $q^* = p/2$.
2. Check the shutdown condition: $\pi(q^*) = p \cdot (p/2) - (16 + (p/2)^2) = p^2/4 - 16$. The firm shuts down if this is less than $\pi(0) = -12$. Solving $p^2/4 - 16 \geq -12$ gives $p \geq 4$.
3. Therefore, the supply function is:

$$s(p) = \begin{cases} 0 & p < 4 \\ 0 \text{ or } 2 & p = 4 \\ \frac{p}{2} & p > 4 \end{cases}$$

Q: What defines the short-run equilibrium price in a competitive market?

A: The price p^* such that market supply equals market demand:

$$S(p^*) = D(p^*)$$

> **Note**
>
> In short-run equilibrium, some firms may have zero or negative profits.

Q: What are the two conditions for long-run equilibrium in perfect competition?

A:

1. The market clears:

$$N \cdot s(p^*) = D(p^*)$$

2. Firms earn zero economic profit (due to free entry and exit):

$$\pi(s(p^*)) = 0 \Leftrightarrow p^* = AC(q^*)$$

Social Welfare

Q: What is the inverse demand function?

A: The **inverse demand function**, $p(q) = D^{-1}(q)$, represents the marginal value to the consumer, $mv(q)$.

Q: What is the total value to consumers for a given quantity q?

A: The total value is the area under the inverse demand curve from 0 to q:

$$v(q) = \int_0^q p(x)dx$$

Q: What is consumer surplus (CS)?

A: Consumer surplus is the total value consumers receive minus what they actually pay:

$$CS = \int_0^q p(x)dx - p(q) \cdot q$$

Q: What is producer surplus (PS)?

A: Producer surplus is the total revenue producers receive minus the total variable cost (the area under the marginal cost curve):

$$PS = p(q) \cdot q - \int_0^q MC(x)dx$$

Q: What is social welfare (SW)?

A: Social welfare is the sum of consumer and producer surplus:

$$SW = CS + PS = \int_0^q p(x)dx - \int_0^q MC(x)dx$$

Tip

Perfect competition maximizes social welfare, resulting in zero deadweight loss ($DWL = 0$).

Monopoly

Q: What is a monopoly?
A: A **monopoly** is a market structure with a single seller.

Q: What are common monopoly pricing instruments?
A:

1. Linear pricing
2. Two-part tariff
3. Price discrimination
4. Bundling

Q: What are the primary causes of monopoly?
A:

1. **Technology:** High fixed costs and a large minimum efficient scale can create a natural monopoly.
2. **Market Size:** Market demand may be too small to support more than one efficient firm.

Q: What is the monopolist's profit function?
A: The profit function is:

$$\pi(q) = R(q) - C(q) = p(q) \cdot q - C(q)$$

Q: What is marginal revenue (MR) for a monopolist?
A: Marginal revenue is the derivative of total revenue with respect to quantity:

$$MR(q) = p(q) + p'(q) \cdot q$$

Q: What is the first-order condition (FOC) for monopoly profit maximization?
A: The firm produces where marginal revenue equals marginal cost:

$$MR(q^m) = MC(q^m)$$

Q: What is the shutdown condition for a monopolist?
A: The monopolist will produce only if:

$$\pi(q^m) \geq \pi(0) \Rightarrow p^m \geq AVC(q^m) + \frac{F - S}{q^m}$$

Q: What is the price elasticity of demand?
A: It measures the responsiveness of quantity demanded to a change in price and is defined as:

$$\epsilon(p) = \frac{p \cdot D'(p)}{D(p)}$$

Q: What is Lerner's Index of Market Power (LIMP)?
A: It measures the proportional markup of price over marginal cost and is inversely related to the price elasticity of demand:

$$\frac{p - MC(q)}{p} = -\frac{1}{\epsilon(q)}$$

> ⚠ Warning
>
> A profit-maximizing monopoly will never operate in the inelastic part of its demand curve (where $-1 < \epsilon < 0$).

Q: Why is a monopoly considered inefficient?
A: It creates a deadweight loss (DWL > 0) by setting a price above marginal cost, which reduces output and consumer surplus compared to a competitive market.

Q: What are potential social benefits of monopoly?
A:

- Innovation incentives (the prospect of monopoly profits can drive R&D).
- Economies of scale in natural monopolies can lead to lower average costs than under competition.

Q: When does a natural monopoly occur?
A: A natural monopoly occurs when costs are **subadditive**, meaning it is cheaper for one firm to produce the total output than for multiple firms:

$$C(q_1 + q_2) \leq C(q_1) + C(q_2)$$

Monopsony

Q: What is a monopsony?
A: A **monopsony** is a market structure with a single buyer.

Q: What is the total cost of buying for a monopsonist?
A: The total cost is:

$$BC(q) = p(q) \cdot q$$

Q: What is the marginal cost of buying (MBC) for a monopsonist?
A: The marginal cost of buying is the derivative of the total cost of buying:

$$MBC(q) = p(q) + p'(q) \cdot q$$

Q: What is the profit function for a monopsonist?
A: The profit function is:

$$\pi(q) = V(q) - BC(q)$$

Q: What is the optimality condition for a profit-maximizing monopsonist?
A: The monopsonist will buy a quantity where the marginal value equals the marginal cost of buying:

$$MV(q^m) = MBC(q^m)$$

> Note
>
> Because $MBC(q) > p(q)$ (the price of the input), a monopsony results in a lower quantity purchased and a positive deadweight loss (DWL).

Case Study: Amazon

Q: What was the core issue in the dispute between Amazon and the publisher Hachette?
A: Amazon was demanding a higher share of book prices from Hachette. When Hachette refused, Amazon used its market power by delaying shipments of Hachette books and raising their prices on its platform.

Q: Does Amazon exercise monopoly power in the form of high consumer prices?

A: Despite its market dominance, Amazon generally keeps prices low for consumers, which limits the direct, traditional abuse of monopoly power through high pricing.